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Demo study: Sony Bravia (Testing of video-commercials) - 9.0

This help center articles offers background information to the downloadable demo study:

(Version 9.0): Sony Bravia Study 9.0

The Sony Bravia demo study was inspired by two published articles by Ohme et al. (2010). In the first of these two articles, the authors compared price-winning commercials for Sony Bravia flatscreen televisions, among which the commercials Balls (2005) and Paint (2006) using EEG-based measurements of approach and withdrawal-related motivation (i.e., frontal alpha asymmetry). In a second article, the authors further discussed the emotionally inducing effect of one particular scene in the commercial Balls – a frog jumping out of a rain pipe.

The original articles focused on frontal alpha asymmetry (FAA), an EEG-based measurement of approach and withdrawal-related motivation - or 'liking' and 'disliking'. However, for the purpose of this demo study, we added a few more technologies.

Methodologies:

- Survey

How do respondents rate the commercials?

- Eye tracking (Smart Eye Aurora)

What is it that respondents look at/pay attention to in the commercials?

- Facial expression analysis (Affectiva AFFDEX)

Are respondents engaged in what they see?

- Electrodermal activity (Biopac Mp150)

Are respondents emotionally aroused by the visual content?

- Electroencephalography (NeuroElectrics Enobio 32)

Are there signs of approach or rather withdrawal-related motivation?

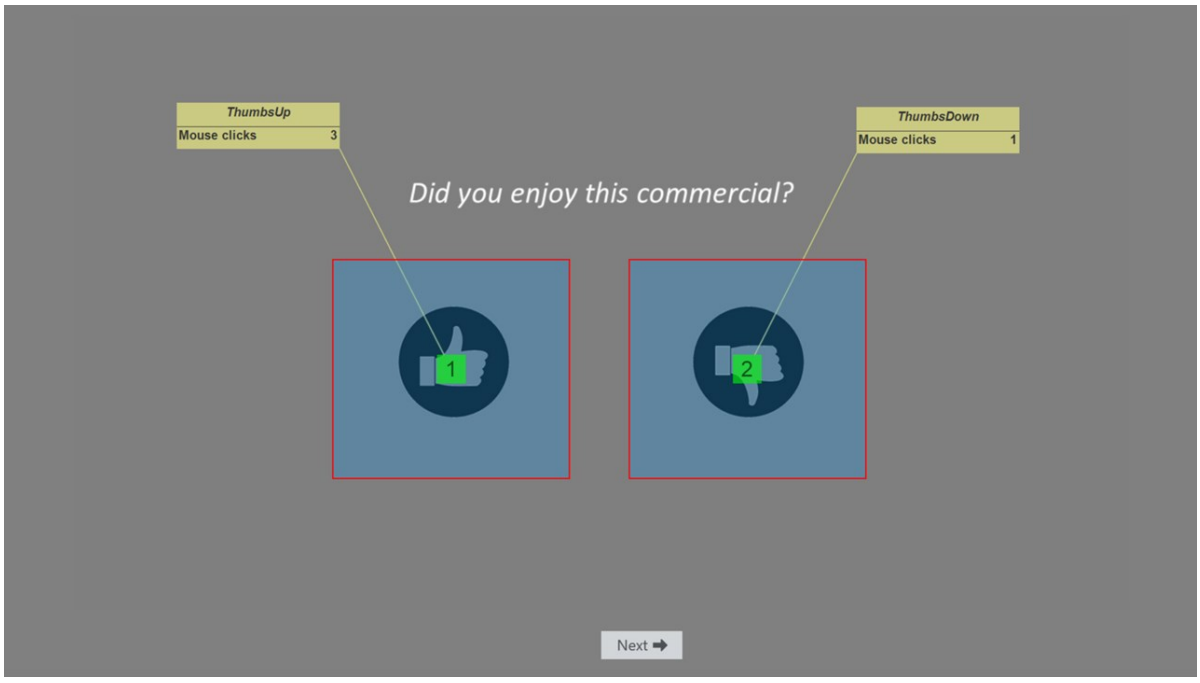
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In the demo study, two (out of four) respondents viewed a version of the Balls commercial *with* the famous frog scene (Commercial 1A), while the other two respondents viewed a version of the same commercial *without* the frog scene (Commercial 1B). All respondents further also viewed the commercial Paint (Commercial 2). As such, the demo study combined a between-subject design and a within-subject design. For the *between*-subject comparison, we compared commercials 1A and 1B; for the *within*-subject comparison, we compared commercials 1 and 2.

Below, we list a few tools for analyzing the data.

Track mouse-clicks to know how the commercials were rated by the respondents

Areas-of-interest (AOIs) can be used to measure mouse-clicks. Note how three (out of four) respondents rated the Paints commercial with a thumbs up, while one (out of four) respondents rated the same commercial with a thumbs down.



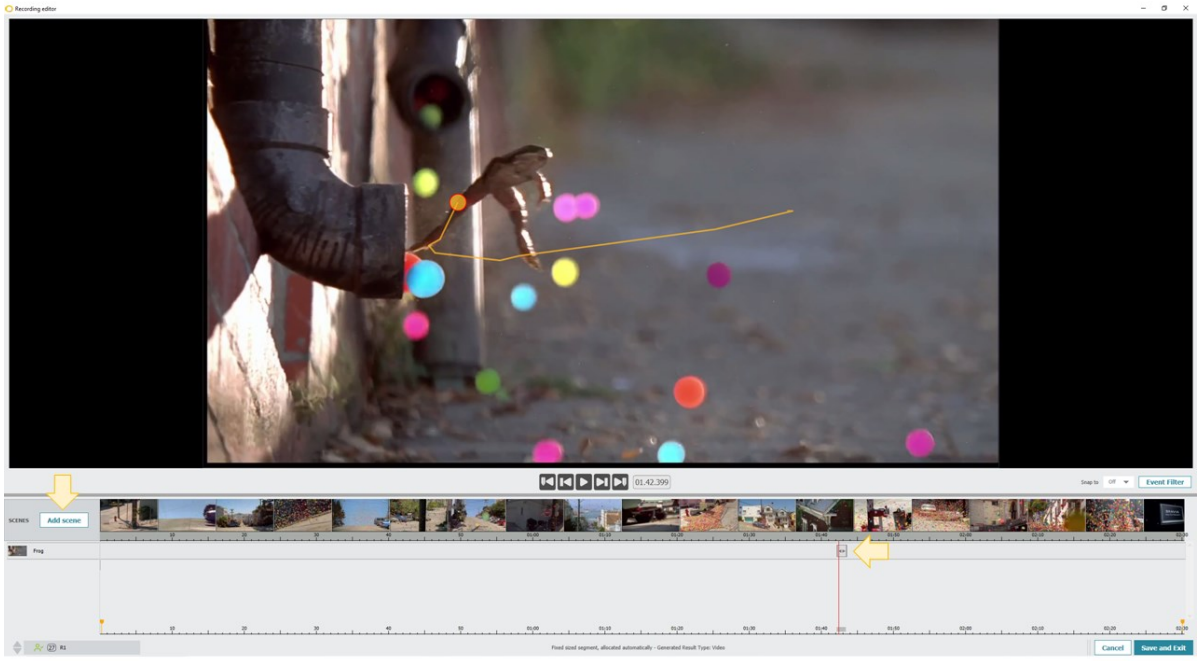
Read more: Visualizing eye tracking data

Read more: Static areas-or-interest templates (AOIs)

Use the scene editor to isolate a scene

The scene editor can be used to isolate the ‘frog’ scene.

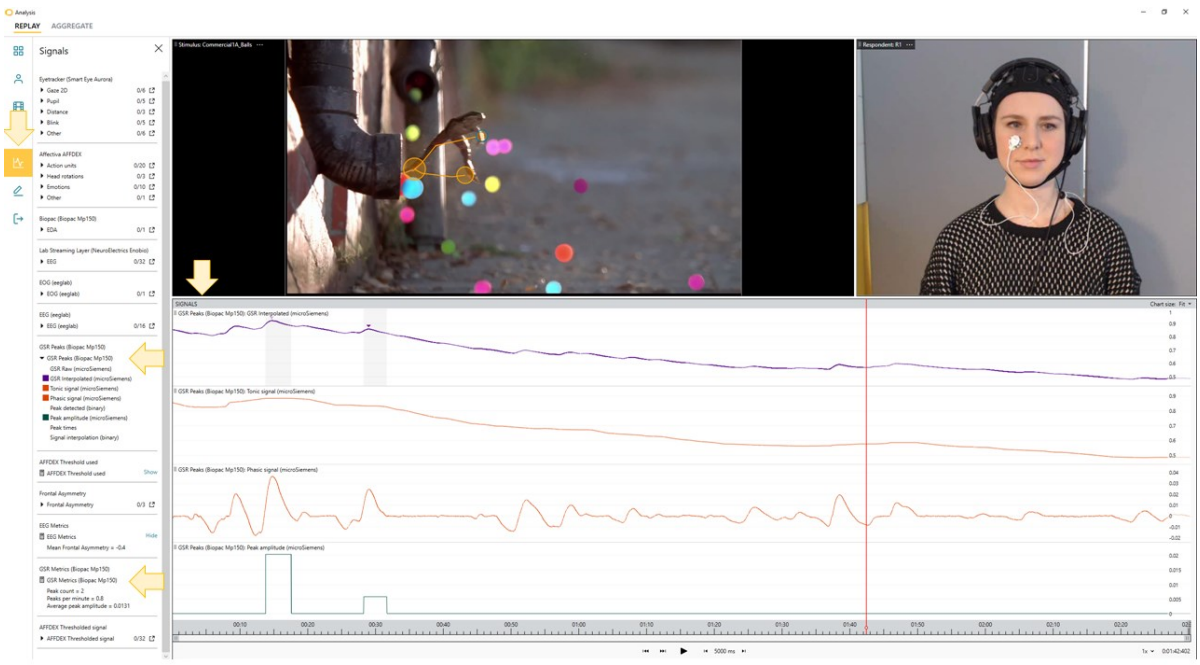
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Read more: How to use the scene editor

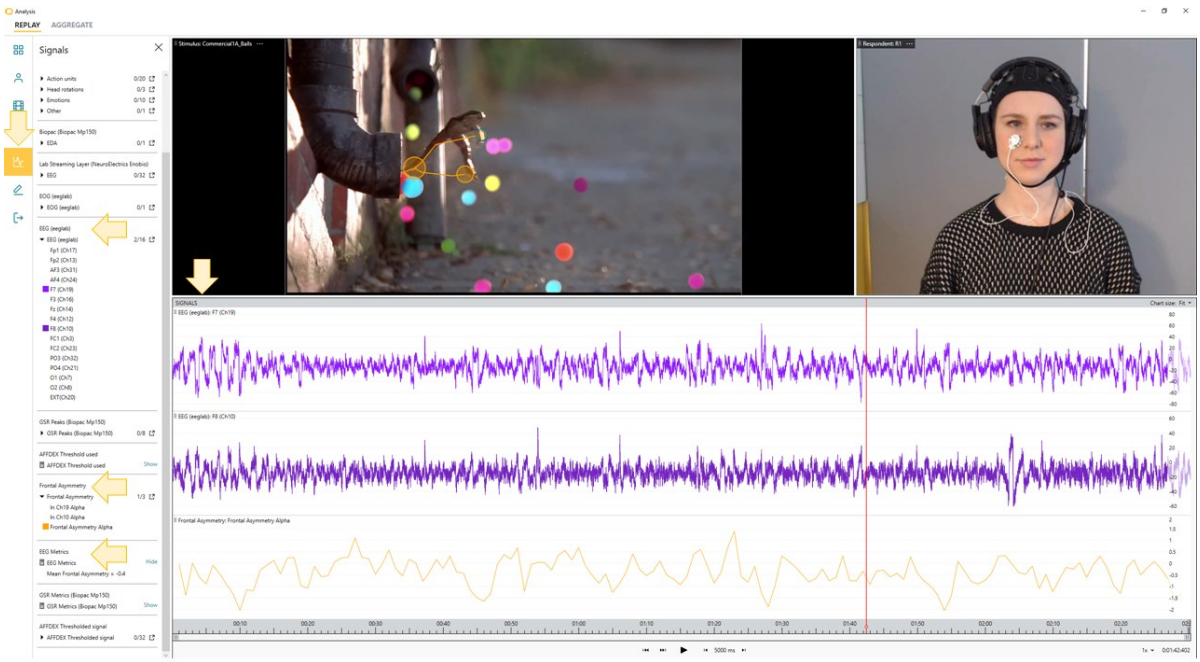
Add R-Notebooks to apply signal processing

The 'GSR peak detection R-Notebook' can be used for measuring the number of peaks in the electrodermal activity. This is a measurement of emotional ‘arousal’.



The 'Frontal alpha asymmetry R-Notebook' can further be used for measuring the difference in alpha power between the left frontal lobe and right frontal lobe. This is an EEG-based measurement of approach (‘liking) and withdrawal-related (‘disliking’) motivation.

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Read more: What are R-Notebooks?

Read more: GSR R-Notebooks: processing in iMotions and algorithms used

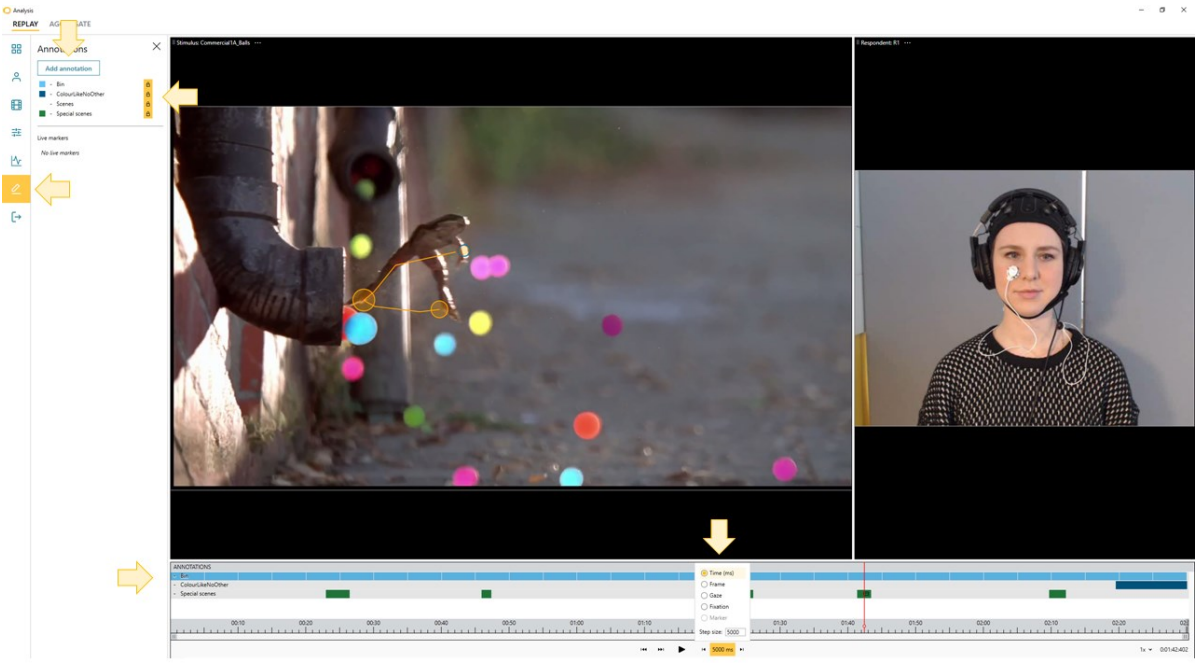
Read more: Frontal alpha asymmetry (FAA): processing in iMotions and algorithms used

Use annotations for binning the data

Binning of the data can be done by defining a 'binning' annotation with a fixed duration (e.g., of 5 seconds).

Alternatively, you may use annotations to mark (special) scenes, for instance, the frog scene or the end-scene ('Colours like no others') used for the analysis below.

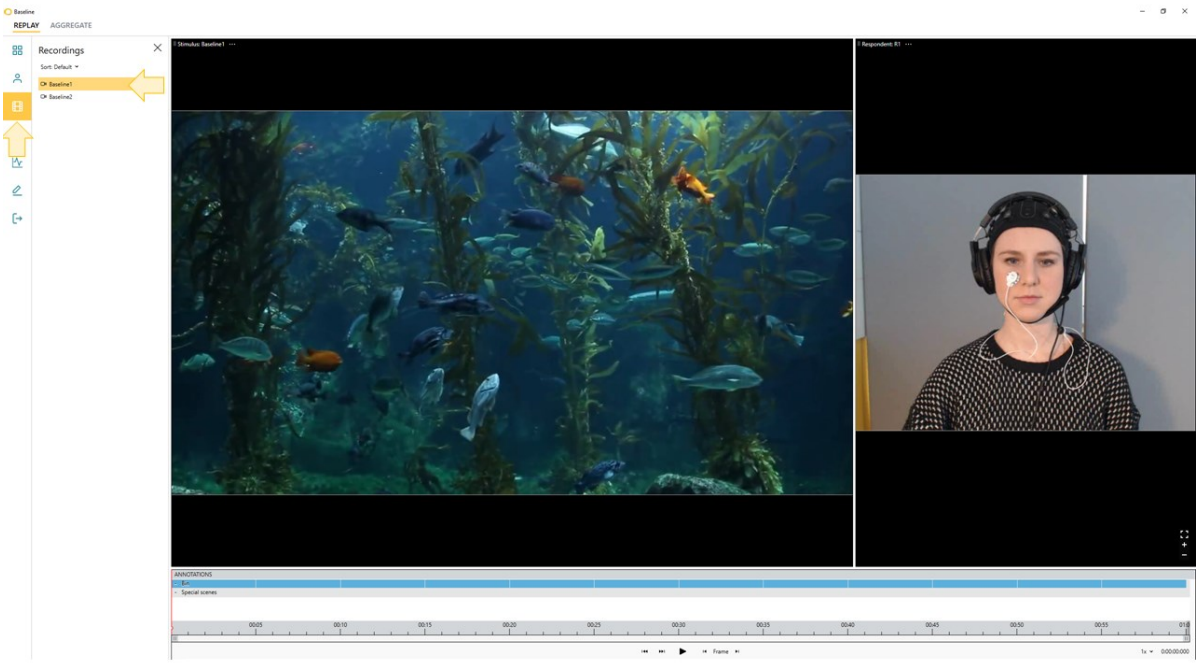
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Read more: Annotations

Compare your results to a baseline for helping the interpretation

Note further that it is often useful to add a 'baseline' to the study to compare your results against – this helps the interpretation of the results. Here, in addition to watching the commercials, the respondents also viewed two 1-minute videos of fishes in a coral reef.

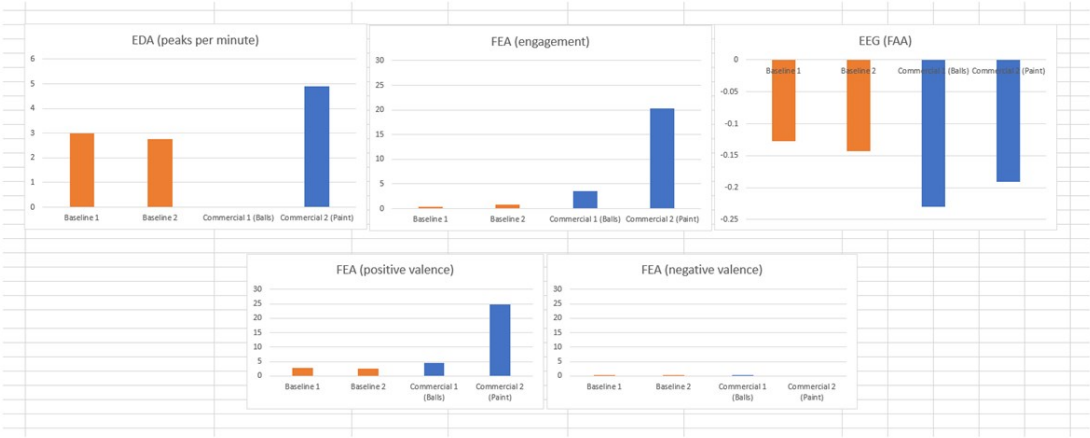


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Example analysis

Below, as an example analysis, we compare the commercials Balls and Paint with respect to (1) the number of peaks in the electrodermal activity (peaks per minute, i.e. a measurement of 'emotional arousal'), (2) the level of engagement revealed by facial expression analysis, and (3) the level of approach or withdrawal-related motivation in the electroencephalography. For this analysis, we focus on the last 10 seconds of the commercial – when the respondents viewed the end-scene with the product slogan 'colours like no other'.

We further contrast these results with the same measurements taken during the baseline periods (1 minute).



What is comforting to observe first, is that the levels of emotional arousal, engagement and approach/withdrawal-related motivation were comparable for the two baseline stimuli (shown in orange). In contrast, the responses for the two commercials Balls and Paints were very different (shown in blue): the commercial Paint was *more* emotionally arousing, *more* engaging and evoked a relatively *stronger* approach motivation (i.e., less negative, hence more positive, frontal alpha asymmetry scores) than the commercial Balls – consistently pointing towards a slight preference of the respondents for the commercial Paint *.

This conclusion would not have been reached by the ratings only: for both commercials three respondents clicked 'thumbs up', while one respondent clicked 'thumbs down'.

*Note that more than four respondents would be needed to draw statistically relevant conclusions; this analysis is merely an example.

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References

Ohme, R., Reykowska, D., Wiener, D., & Choromanska, A. (2010). Application of frontal EEG asymmetry to advertising research. Journal of economic psychology, 31(5), 785-793.

Ohme, R., Matukin, M., & Szczurko, T. (2010). Neurophysiology uncovers secrets of TV commercials. Der markt, 49(3-4), 133-142.

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